

My Way is not Very Sportsmanlike

Shaping Adversary Behavior to Strengthen Defenses

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David J. Bianco

- 20+ years of experience in incident detection & response, threat hunting, CTI, and other Blue-team topics
- SANS Certified Instructor
- Pyramid of Pain & PEAK
- Terrible at introducing himself

splunk>

Some assumptions to start off with

Actors have goals

Threat actors want something

Defenders want to keep them from achieving something

Actors are (mostly) rational

Both sides act in their own best interests, barring mistakes

Actions tend to further progress toward goals

Some are less rational than others

Actors respond to actions

Both sides make moves & both sides respond to moves by the other.

Defenders always get the first move(s)

Using our **first move**
advantage, defenders set
the pace, tone, and style
of engagement.



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You've heard of the Defender's Dillema: "Defenders have to get it right every time. Attackers only have to get it right once."

Did you know this is a lie?

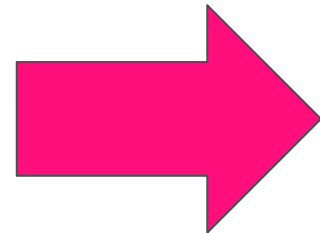
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12:55 PM · May 24, 2021 · Twitter Web App



SAFETY & SECURITY

How AI can strengthen digital security



Feb 16, 2024

5 min read

We're launching the AI Cyber Defense Initiative to help transform cybersecurity and use AI to reverse the dynamic known as the "Defender's Dilemma"



Phil Venables

Vice President, Chief Information Security Officer (CISO), Google Cloud



Royal Hansen

Vice President of Engineering for Privacy, Safety, and Security

 Share

How to tilt the cybersecurity balance from attackers to defenders

Today, and for decades, the main challenge in cybersecurity has been that attackers need just one successful, novel threat to break through the best defenses. Defenders, meanwhile, need to deploy the best defenses at all times, across increasingly complex digital terrain — and there's no margin for error. This is the "Defender's Dilemma," and there's never been a reliable way to tip that balance.

The Defender's Dilemma is actively harmful

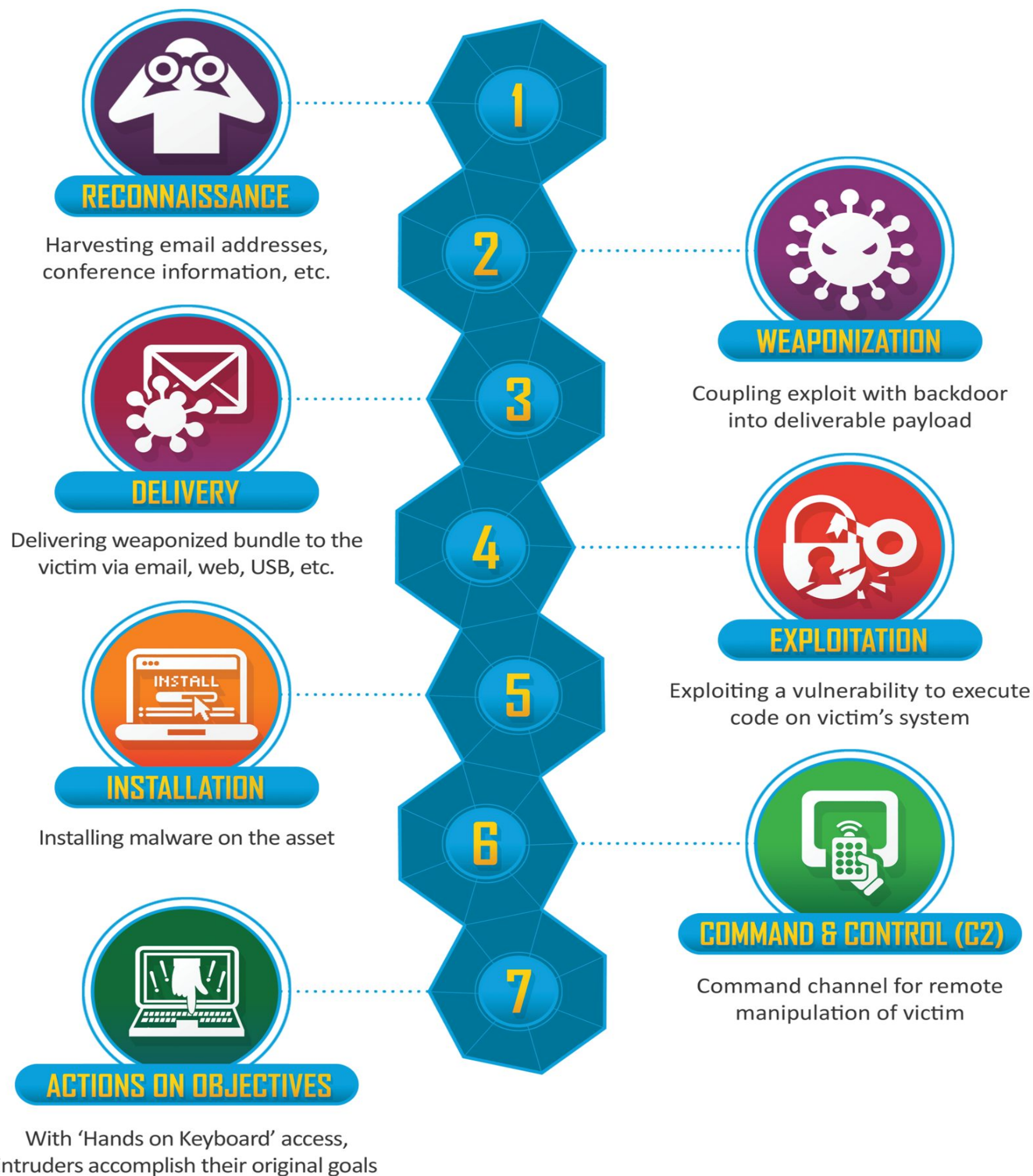
Wrong Premise

Bad Decisions

Wasted
Resources

Demoralized
Defenders

Bad Security
Outcomes



Attacks are not points in time

Attacks are points **over** time

A phish or an exploit is **only** a **foothold**.

Other phases come both **before** and **after**.

We can:

- **Detect** at (almost) any stage
- **Investigate** to cover all stages
- **Interdict** before they reach the final stage

Attackers have to get it
right through their **entire**
attack.

Defenders only need to
detect them **once.**

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Nail the basics

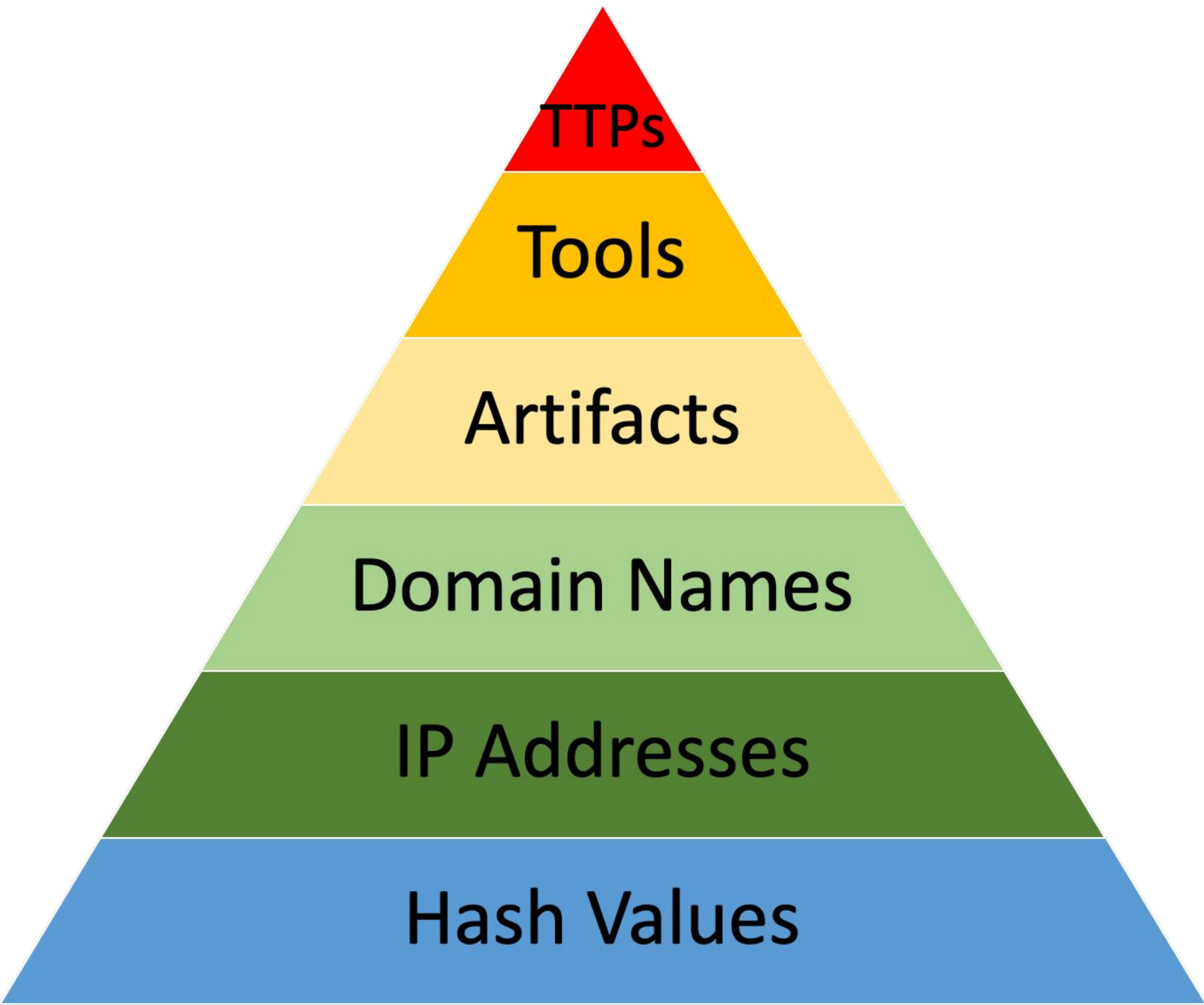
Eat your cyber-vegetables

None of these are exciting.

All are important.

- Asset & configuration management
- Software updates & patching
- Multi-factor authentication
- Endpoint security
- Network monitoring
- Security awareness





Impose Cost

Practice “offensive defense”

The **Pyramid of Pain** is very popular, of course.

Resilient systems are much more difficult to manipulate.

Anything **proactive** (e.g., threat hunting)

Maybe you can think of some things in your own environments

<https://bit.ly/PyramidOfPain>

Lie

Cheat your assets off

There's **no such thing as cheating** for the Blue Team.

Attackers have **needs, goals, and habits**. Take advantage of them.

Deception technology makes it quite easy to **lie and cheat at scale**.



Conclusion/Wrap-up

A “disadvantage mindset” starts us off on the wrong foot, leads to burnout and bad security outcomes

For best defense, look for and exploit the many opportunities we have based on our first move advantages.

Both attacker & defender can influence the other’s behavior. It’s up to us to maximize our ability to drive attackers into behaviors which are beneficial to us.

(And it doesn’t have to cost a lot)

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